Module 7, Lesson 3
AI 101 for Mechanical Engineers
A Minimal, Honest Regression Workflow

Undergraduate Engineering Course

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Outline

1. Fundamentals of Regression
2. The Evaluation Workflow
3. Application and Limitations
4. Conclusion

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01

1. Fundamentals of Regression

Define core mathematical evaluation metrics, standard terminology, and fundamental error units.

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Metric Diagnostic Table

Metric	Mathematical Definition	Example SI Unit	Interpretation
Linear Regression	Model linear in fitted coefficients	Derived from inputs	
MAE	Average $\lvert y - \hat{y} \rvert$	kW	Mean absolute prediction error
RMSE	$\sqrt{(\text{mean}((y - \hat{y})^2)}$	kW	Large errors contribute strongly
R^2	$1 - \frac{(\text{squared error}}{\text{variance})}$	Dimensionless (Relative)	Not a physical error in Watts

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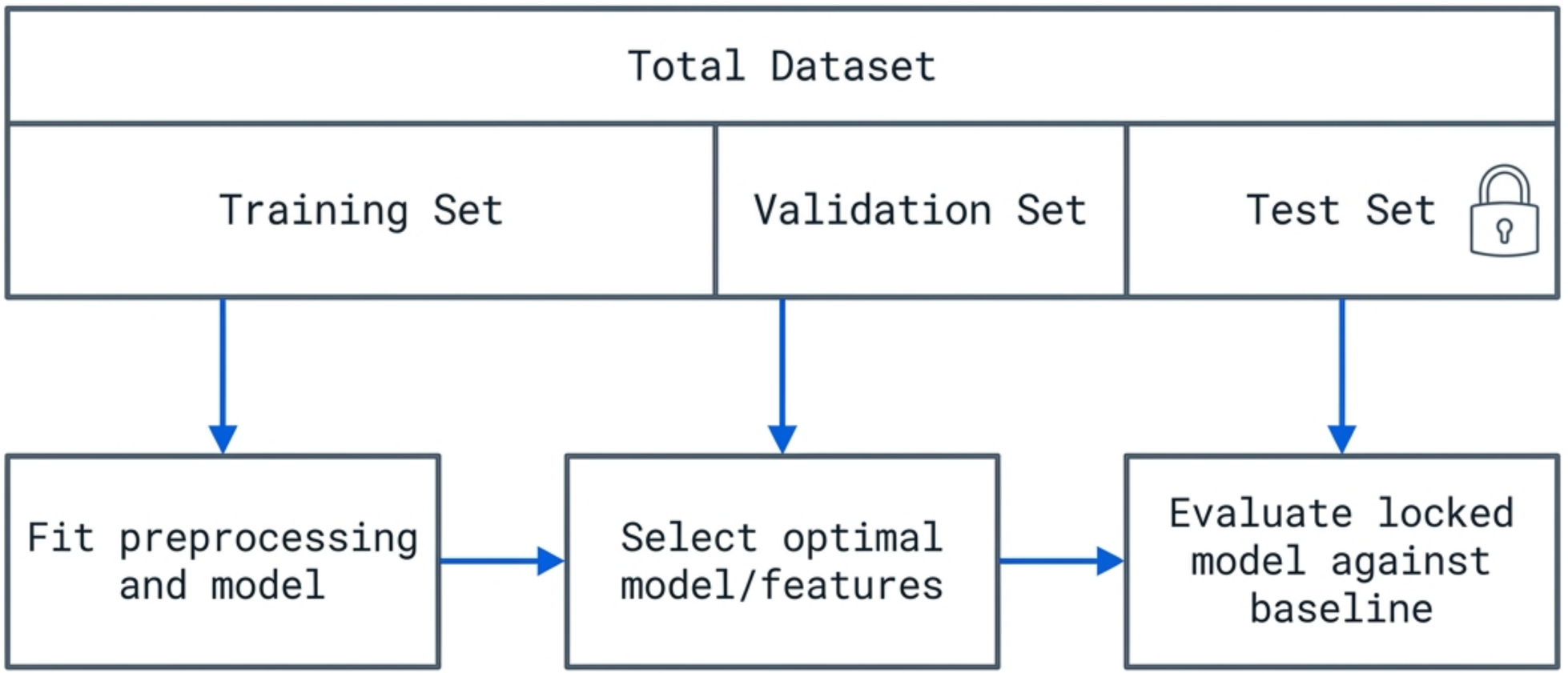
02

2. The Evaluation Workflow

Establish the rigid chronological sequence and evidence constraints required for valid model evaluation.

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Pipeline Sequence Schematic



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Evidence Utilization Boundaries

Workflow Stage	Allowed Evidence Application
Training	Fit model parameters and learned preprocessing.
Validation	Select among model configurations or feature choices.
Locked Final Test	Estimate final performance after selection only.

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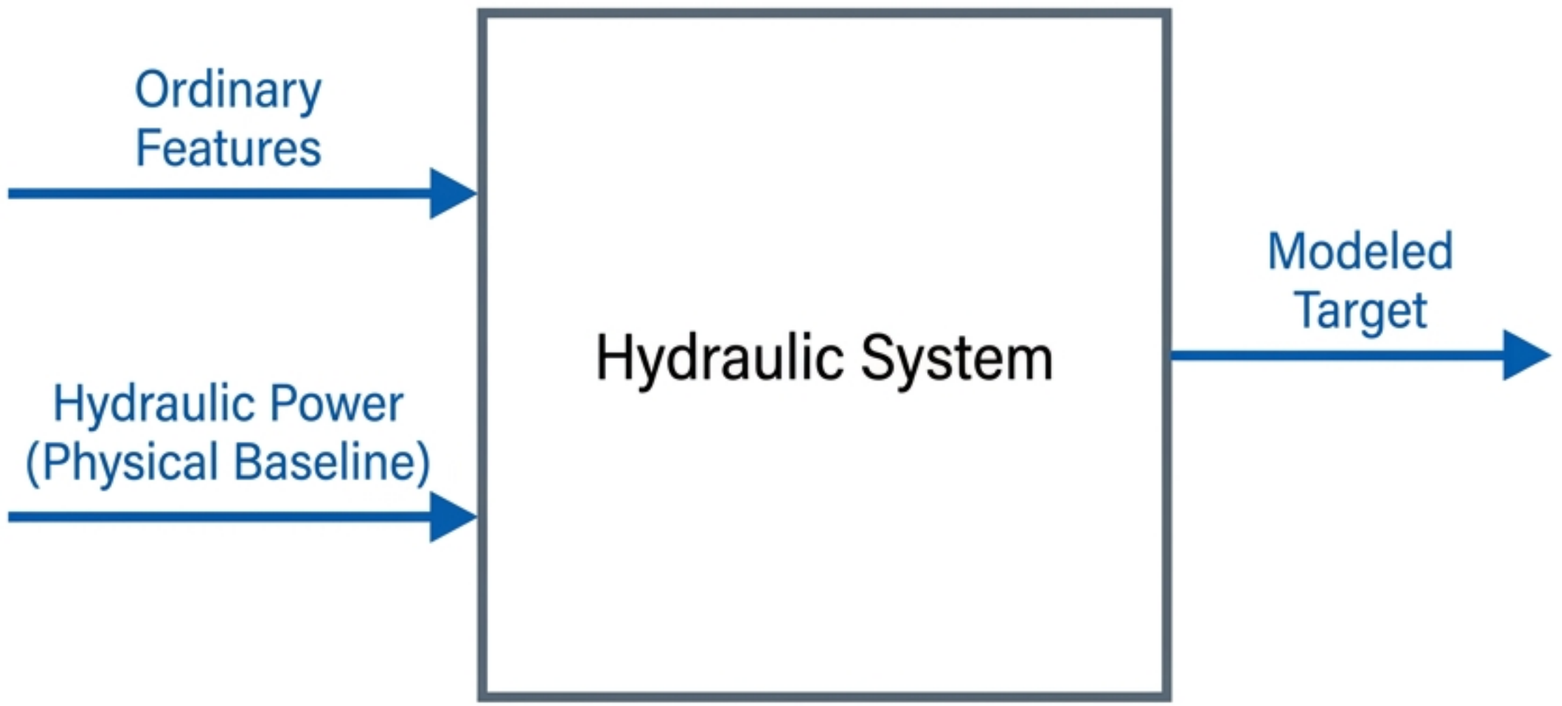
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3. Application and Limitations

Apply workflow principles to physical system system data and synthesize the requirements honest engineering reports.

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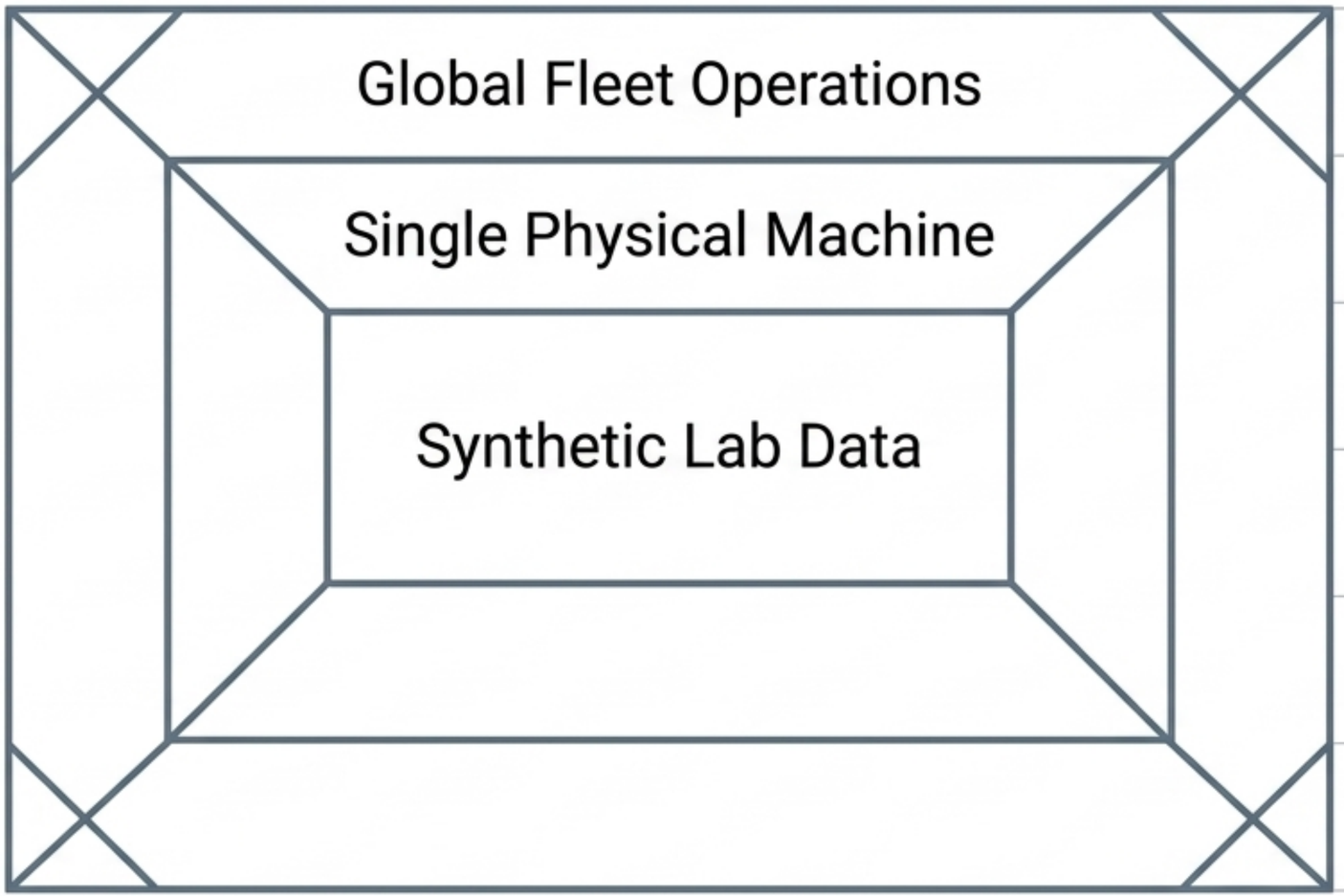
Physical System Data Flow



- Independent synthetic designs utilize random data splitting.
- Chronological forecasting requires sequential time-based splitting.
- Final test metrics are reported only after validation.

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System Generalization Constraints



- Synthetic evidence cannot validate physical equipment safety.
- Single-machine models prevent automatic fleet-wide transfer.
- Evaluated load ranges restrict valid operational limits.

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Sound Regression Reporting Requirements

- ☑ State evaluated physical target units.
- ☑ Detail data provenance and split strategy.
- ☑ Compare against a specific baseline performance.
- ☑ Define explicit operating-domain boundary limits.

Constraint: A high R^2 score alone cannot substantiate a global safety claim.

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4. Conclusion

Summarize the definitive principles of a minimal, honest regression workflow.

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Key Points Summary

[01] Fundamentals: MAE and RMSE quantify dimensional error; R^2 provides dimensionless relative scoring.

[02] Workflow: Data splitting enforces strict evaluation boundaries for training, validation, and testing.

[03] Application: Honest evaluation mandates physical baselines, declared SI units, and explicit generalization limits.

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